

L12 Stochastic Games (Markov Decision Processes).

CS 280 Algorithmic Game Theory

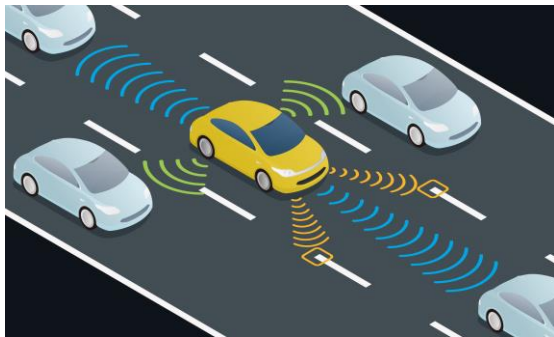
Ioannis Panageas

Multi-agent systems and RL

Decentralized systems

Individual interests (rational agents, cooperation/competition etc)

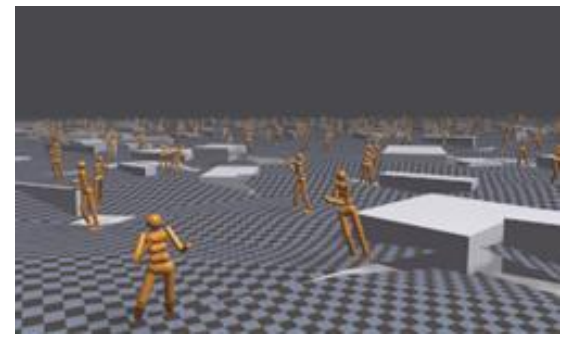
Distributed optimization



Self-driving cars



Auctions



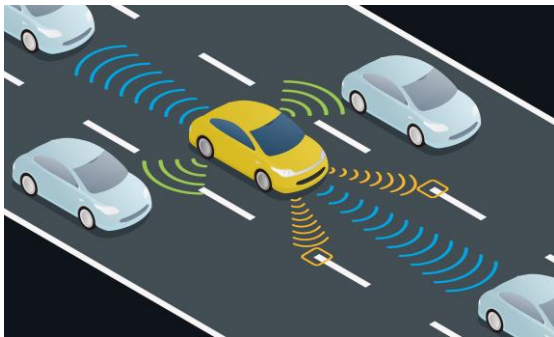
Robotics

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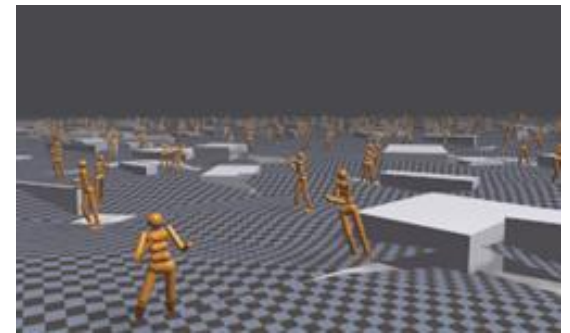
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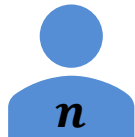


Robotics

How these systems evolve? Predictions?

Markov Games

Markov games or *stochastic* games are established as a framework for multi-agent reinforcement learning [Littman, 1994].



n number of players

Markov Games

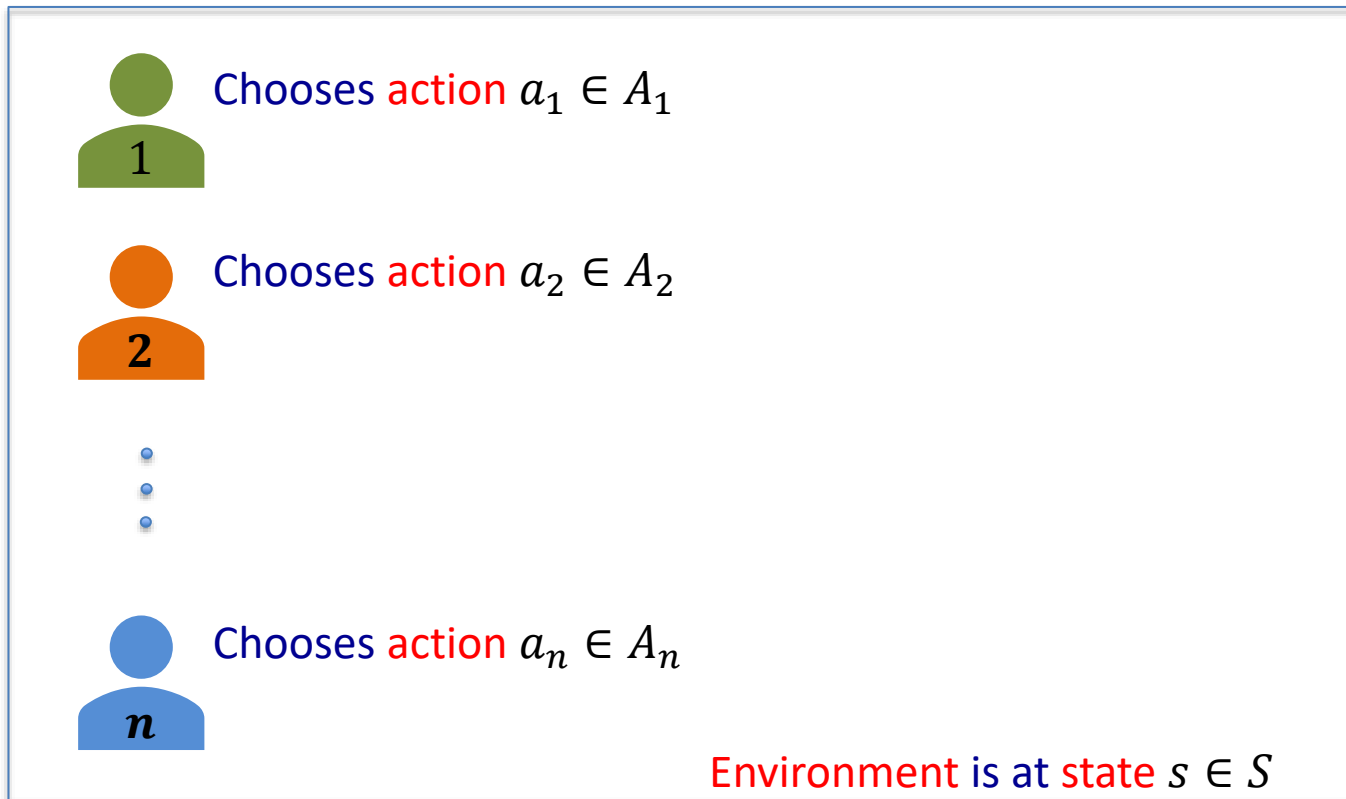
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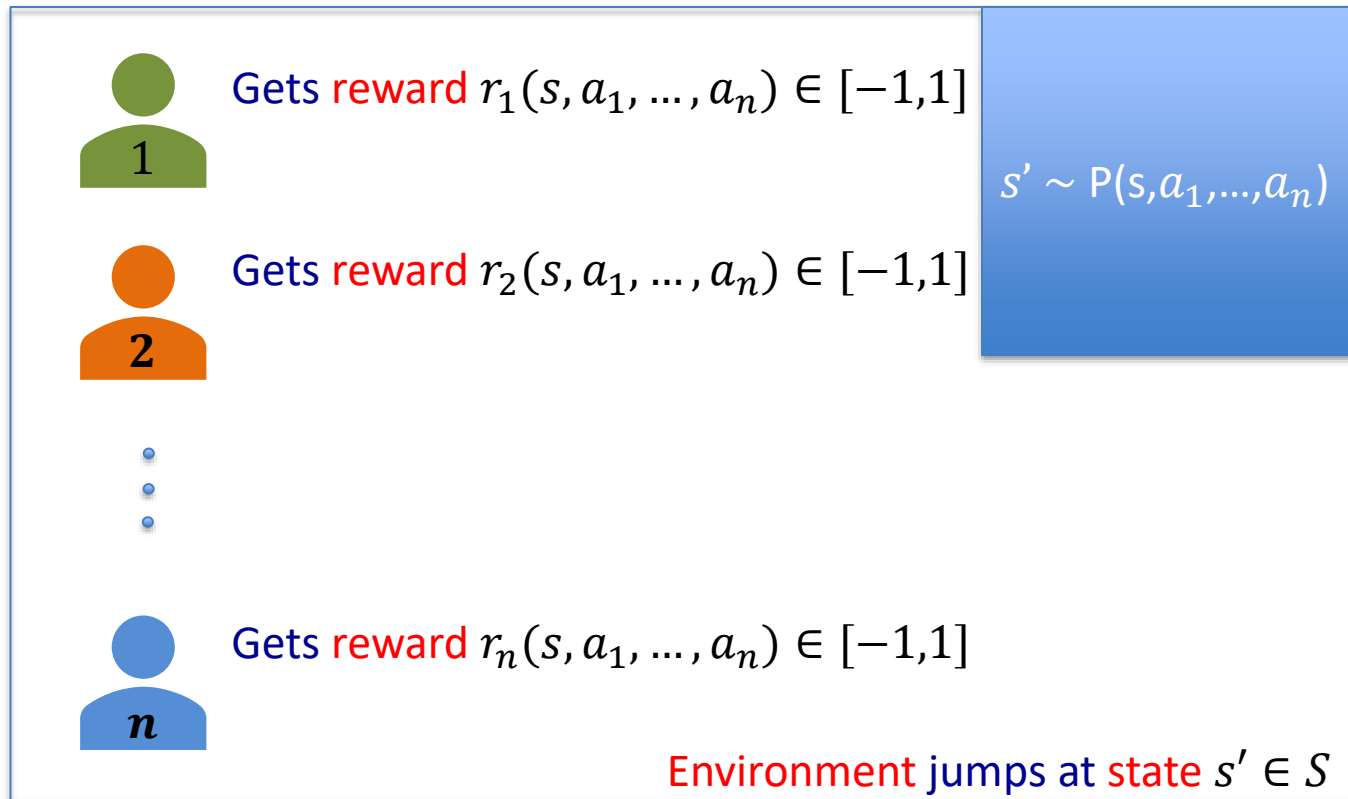
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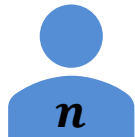
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Gets **value** $V_1(s^0) := \sum_{t=0}^H r_1(s^t, a_1^t, \dots, a_n^t)$



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n number of **players**

Markov Games

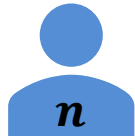
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If H is ∞ , then we introduce a discount γ

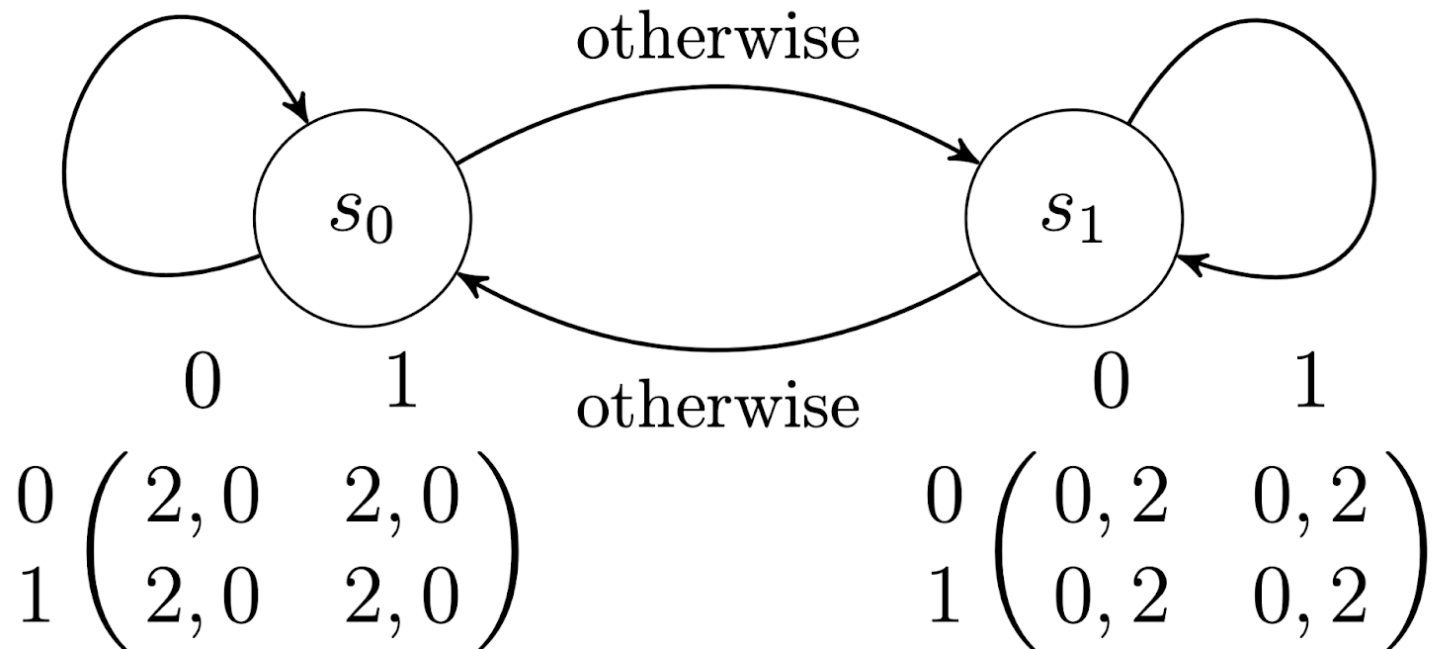
e.g., $V_1(s^0) := \sum_{t=0}^{\infty} \gamma^t r_1(s^t, a_1^t, \dots, a_n^t)$

n number of players

An example

$$a_A^0 \oplus a_B^0 = 0$$

$$a_A^1 \oplus a_B^1 = 0$$



n -player Markov game: Formal definition

Markov games or *stochastic* games are established as a framework for multi-agent reinforcement learning [Littman, 1994]

- $\mathcal{N}$, a finite set of agents with $n := |\mathcal{N}|$,
- $\mathcal{S}$, a finite state space,
- $\mathcal{A}_k$, a finite action space each player k , and $\mathcal{A} = \times_{k=1}^n \mathcal{A}_k$,
- $r_k : \mathcal{S} \times \mathcal{A} \rightarrow [-1, 1]$, a reward function for each agent k ,
- $P : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ a transition probability function,
- $\gamma \in [0, 1)$, a discount factor,
- $p \in \Delta(\mathcal{S})$, an initial state distribution.

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- *Single agent RL*

The framework

A finite Markov Decision Process (MDP) is defined as follows:

- A finite state space $\mathcal{S}$.
- A finite action space $\mathcal{A}$.
- A transition model $\mathbb{P}$ where $\mathbb{P}(s'|s, a)$ is the probability of transitioning into state s' upon taking action a in state s . $\mathbb{P}$ is a matrix of size $(S \cdot A) \times S$.
- Reward function $r : \mathcal{S} \times \mathcal{A} \rightarrow [-1, 1]$.
- A discounted factor $\gamma \in [0, 1)$.
- $\boldsymbol{\rho} \in \Delta(\mathcal{S})$, an initial state distribution.

Definitions

Definition (**Markovian stationary policy**). *Policy is called a function*

$$\pi : \mathcal{S} \rightarrow \mathcal{A}.$$

Definition (**Value function**). *Given a policy π the value function is given by*

$$V^\pi(\boldsymbol{\rho}) = \mathbb{E}_{\pi, \mathbb{P}} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 \sim \boldsymbol{\rho} \right]$$

The **goal** is to solve

$$\max_{\pi} V^\pi(\boldsymbol{\rho}).$$

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Remarks

- The **max** operator is over all (possibly non-stationary and randomized) policies.
- It suffices to focus on **deterministic**.
- V is **not concave** in π .

Example

Example (Navigation). Suppose you are given a *grid map*. The state of the agent is their *current location*. The four *actions* might be moving 1 step along each of east, west, north or south. The transitions in the simplest setting are deterministic. There is a goal g that is trying to reach. *Reward* is one if the agent reaches the goal and zero otherwise.

0.729	0.81	0.9	★
0.656		0.81	0.9
0.590	0.656	0.729	0.81

→	→	→	★
↑		↑	↑
↑	→	↑	↑

Remark

- What is V ?
- What is γ in the example?

Bellman operator

Definition (Bellman Operator). *Let's define the following operator $\mathcal{T}$:*

$$\mathcal{T} W(s) = \max_{a \in \mathcal{A}} \{ r(s, a) + \gamma \sum_{s'} \mathbb{P}(s' | s, a) W(s') \}$$

Set $V^*(s) := \max_{\pi} V^{\pi}(s)$.

Claim (Bellman Operator). *V^* is the unique fixed point of the operator.*

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$$\|\mathcal{T}V - \mathcal{T}V'\|_{\infty} = \left\| \max_a \{ r(s, a) + \gamma \sum_{s'} \mathbb{P}(s' | a, s) V(s') \} - \max_{a'} \{ r(s, a') + \gamma \sum_{s'} \mathbb{P}(s' | a', s) V'(s') \} \right\|_{\infty}.$$

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Bellman operator

$$\|x - y\|_\infty \geq \left| \|x\|_\infty - \|y\|_\infty \right|$$

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Bellman operator

$$\|Ax\|_\infty \leq \|A\|_\infty \|x\|_\infty$$

$$\begin{aligned}\|\mathcal{T}V - \mathcal{T}V'\|_\infty &= \left\| \max_a \{r(s, a) + \gamma \sum_{s'} \mathbb{P}(s'|a, s) V(s')\} - \max_{a'} \{r(s, a') + \gamma \sum_{s'} \mathbb{P}(s'|a', s) V'(s')\} \right\|_\infty \\ &\leq \left\| \max_a \{r(s, a) + \gamma \sum_{s'} \mathbb{P}(s'|a, s) V(s') - r(s, a) - \gamma \sum_{s'} \mathbb{P}(s'|a, s) V'(s')\} \right\|_\infty \\ &= \gamma \left\| \max_a \{\mathbb{P}_a(V - V')\} \right\|_\infty \\ &\leq \gamma \|V - V'\|_\infty \quad \text{since } \|\mathbb{P}_a\|_\infty = 1.\end{aligned}$$

Remarks

- Bellman operator is contracting for **infinity** norm.
- Applying the operator **does not give a polynomial time** algorithm. Why?
- **Linear programming** can give optimal policies in polynomial time.

Value Iteration

Idea: We build a sequence of value functions. Let V_0 be any vector, then iterate the application of the optimal Bellman operator so that given V_k at iteration k we compute

$$V_{k+1} = TV_k.$$

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The policy will be given at every iteration as

$$\pi_k = \arg \max_a (1 - \gamma)r(s, a) + \gamma \sum_{s'} P(s'|s, a) V_k(s')$$

After $k = \frac{\log(1/\epsilon)}{\log(1/\gamma)}$ we have error ϵ .

Policy Iteration

Idea: We build a sequence of policies. Let π_0 be any stationary policy. At each iteration k we perform the two following steps:

1. **Policy evaluation** given π_k , compute V^{π_k} .
2. **Policy improvement**: we compute the *greedy* policy π_{k+1} from V^{π_k} as:

$$\pi_{k+1}(x) \in \arg \max_{a \in A} \left[r(x, a) + \gamma \sum_y p(y|x, a) V^{\pi_k}(y) \right].$$

The iterations continue until $V^{\pi_k} = V^{\pi_{k+1}}$.

- *Markov games: Solution concepts*

Solution Concept: Nash equilibrium

- Every agent k picks a policy π_k : 4 possibilities
 1. Markovian and stationary.
 2. Markovian and non-stationary.
 3. Non-Markovian and stationary.
 4. Non-Markovian and non-stationary.

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An ϵ -approximate *Nash equilibrium (NE)* $\pi^* = (\pi_1^*, \dots, \pi_n^*)$ means that no agent can unilaterally increase their expected value more than ϵ ,

$$V_k^{\pi^*}(\boldsymbol{\rho}) \geq V_k^{(\pi'_k, \pi_{-k}^*)}(\boldsymbol{\rho}) - \epsilon, \quad \forall k \in \mathcal{N}, \forall \pi'_k.$$

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- Agents **do not share** randomness.
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Remarks

- Agents **do not share** randomness.
- Fixing all agents but i , induces a **classic MDP**. Every agent aims at (approximate) **best response**.
- Generalizes notion of **Nash Equilibrium**.
- Nash policies **always exist** (Fink 64).

The bad news

- Markov games **generalize** normal form games.



Inherit *computational* intractability

The bad news

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Inherit **computational** intractability

[Daskalakis, Goldberg, Papadimitriou 06]

[Chen, Deng 06]

[Rubinstein 15]

PPAD-hard

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PPAD-hard

Specific classes of games?

- *Two-player zero sum
Markov games*

2-player zero-sum Markov games

- $\mathcal{N} = \{1, 2\}$, i.e., $n = 2$,
- $\mathcal{A}, \mathcal{B}$, the finite action space of players 1, 2 respectively.
- $r_2 = -r_1$,
- rest the same.

Conventions

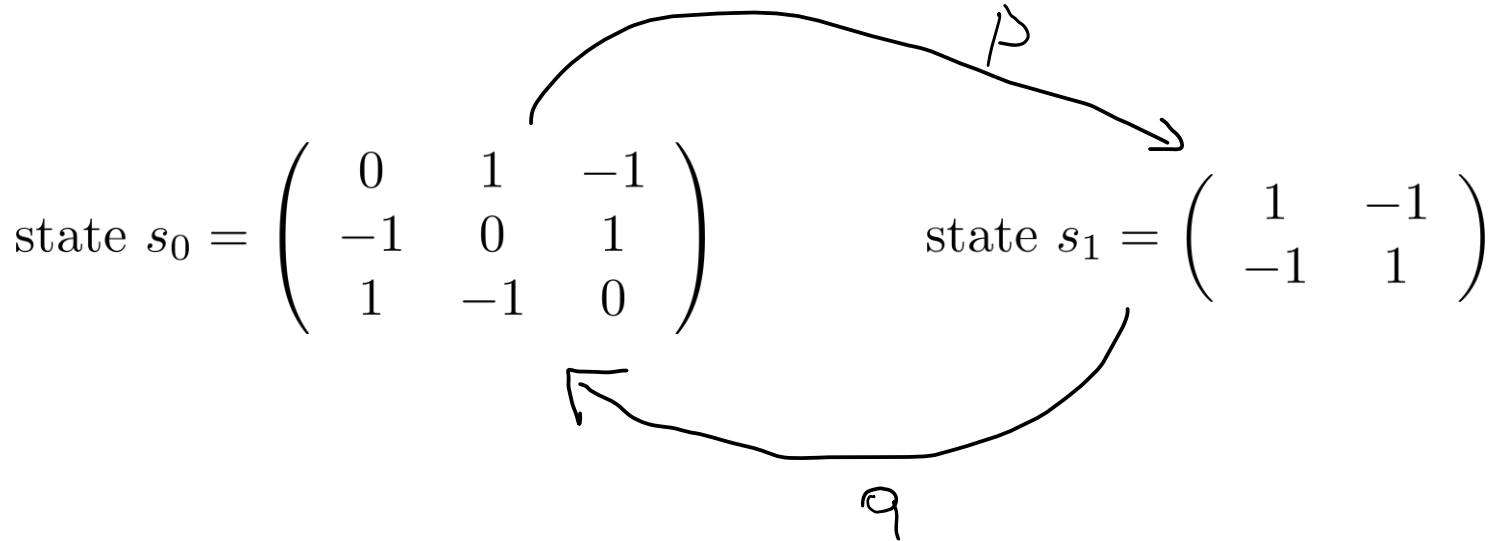
- We call player 2 the maximizer and player 1 the minimizer.
- The value of maximizer is $V^{(\pi_1, \pi_2)}(\rho)$.

2-player zero-sum Markov games

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2-player zero-sum Markov games

A crucial property:

Theorem (Shapley 53). *In any two-player zero-sum Markov game*

$$\min_{\pi_1} \max_{\pi_2} V^{\pi_1, \pi_2}(\boldsymbol{\rho}) = \max_{\pi_2} \min_{\pi_1} V^{\pi_1, \pi_2}(\boldsymbol{\rho})$$

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Remark

- The game has a unique value V^* (recall Von Neumann for normal form two player zero-sum games).
- The theorem implies it does not matter who plays first.
- The function is **not convex-concave!**
- The proof of Shapley uses a **contraction** argument.
- The complexity of finding a Nash equilibrium is *unknown*.

2-player zero-sum Markov games

Proof. Similar to Bellman, **different operator**.

Let $\text{val}(\cdot)$ be the operator applied to a payoff matrix that returns the value of the corresponding zero-sum game.

$$\text{e.g., } \text{val}\left(\begin{bmatrix} -1, 1 \\ 1, -1 \end{bmatrix}\right) = 0.$$

2-player zero-sum Markov games

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$$\text{e.g., } \text{val}\left(\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\right) = 0.$$

Fact: $|\text{val}(A) - \text{val}(B)| \leq \max_{i,j} |A_{ij} - B_{ij}|$

Given a value vector $V(s)$, we define the operator $\mathcal{T}$

$$\mathcal{T}V(s) := \text{val}(r_2(s, \cdot, \cdot) + \gamma \sum_{s'} \mathbb{P}(s' | s, \cdot, \cdot) V(s')).$$

2-player zero-sum Markov games

$$\begin{aligned}\|\mathcal{T}V - \mathcal{T}V'\|_\infty &= \left\| \text{val}\{r(s, \cdot, \cdot) + \gamma \sum_{s'} \mathbb{P}(s'|s, \cdot, \cdot) V(s')\} - \text{val}\{r(s, \cdot, \cdot) + \gamma \sum_{s'} \mathbb{P}(s'|s, \cdot, \cdot) V'(s')\} \right\|_\infty \\ &\leq \left\| \max_{a,b} \{r(s, a, b) + \gamma \sum_{s'} \mathbb{P}(s'|s, a, b) V(s') - r(s, a, b) - \gamma \sum_{s'} \mathbb{P}(s'|s, a, b) V'(s')\} \right\|_\infty \\ &= \gamma \left\| \max_{a,b} \{\mathbb{P}_{a,b}(V - V')\} \right\|_\infty \\ &\leq \gamma \|V - V'\|_\infty\end{aligned}$$

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$$\begin{aligned}\|\mathcal{T}V - \mathcal{T}V'\|_\infty &= \left\| \text{val}\{r(s, \cdot, \cdot) + \gamma \sum_{s'} \mathbb{P}(s'|s, \cdot, \cdot) V(s')\} - \text{val}\{r(s, \cdot, \cdot) + \gamma \sum_{s'} \mathbb{P}(s'|s, \cdot, \cdot) V'(s')\} \right\|_\infty \\ &\leq \left\| \max_{a,b} \{r(s, a, b) + \gamma \sum_{s'} \mathbb{P}(s'|s, a, b) V(s') - r(s, a, b) - \gamma \sum_{s'} \mathbb{P}(s'|s, a, b) V'(s')\} \right\|_\infty \\ &= \gamma \left\| \max_{a,b} \{ \mathbb{P}_{a,b}(V - V') \} \right\|_\infty \\ &\leq \gamma \|V - V'\|_\infty\end{aligned}$$

Remarks

- Bellman operator is contracting for **infinity** norm.
- Applying the operator **does not give a polynomial time** algorithm. Why?

Policy Gradient Iteration

Definition (**Direct Parametrization**). *Every agent uses the following:*

$$\pi_k(a \mid s) = x_{k,s,a}$$

with $x_{k,s,a} \geq 0$ and $\sum_{a \in A_k} x_{k,s,a} = 1$.

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$$x_k^{(t+1)} := \Pi_{\Delta(A_k)^S}(x_k^{(t)} + \eta \nabla_{x_k} V_k^{x^{(t)}}(\rho)),$$

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Some facts about Policy Gradient

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Theorem (Policy Gradient Ascent [Agarwal et al 2020]). *It can be shown for one agent that after $O(1/\epsilon^2)$ iterations, an ϵ -optimal policy can be reached.*

Theorem (Policy Gradient Descent/Ascent [Daskalakis et al 2020]). *It can be shown a two-time scale Policy Gradient Descent/Ascent can give an ϵ -Nash equilibrium in $\text{poly}(1/\epsilon)$ time.*

Remarks

- No guarantees for more than **two** players (only very specific settings).
- Can we find other **classes** of Markov games that PGA converges?
- In general, approximating even stationary CCE is PPAD-complete [Daskalakis et al 2022].